

# **Personalizing Vague Text-to-Music Generation Prompts by Leveraging User Listening Histories**

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## ABSTRACT

This thesis investigates whether vague text-to-music generation prompts can be personalized using PMI-weighted tag co-occurrence patterns informed by user listening history metadata.<sup>1</sup> A combined dataset was created from the Million Song Dataset, Taste Profile, and Last.fm tags. This dataset contains 50 users and includes the titles of the songs they have listened to, high-level track features (tempo, key, and mode), as well as corresponding human-annotated semantic tags gathered from the Last.fm dataset. Based on this dataset, 10 vague prompt candidates were identified and tested.

The augmentation of these vague prompts is addressed using two methods: **(1)** PMI-weighted co-occurrence patterns computed across all users’ listening histories, reflecting the collective understanding of a vague prompt, and **(2)** PMI-weighted co-occurrence patterns computed from a single user’s listening history, addressing a personal understanding of the vague music prompt. To evaluate these methods in the context of automatic personalization for music generation, MusicGen-Large was used to generate 10 second music clips under three prompt conditions: **(1)** unaugmented, **(2)** global augmentation, and **(3)** personal augmentation.

Evaluation was conducted objectively using Fréchet Audio Distance (FAD) and CLAP text-to-audio similarity measures. The experimental results suggest that both augmentation strategies improve upon the unaugmented baseline and produce music that is distributionally closer to user-specific held-out reference audio. However, personal augmentation did not produce statistically significant improvements over global augmentation overall. Per-prompt analysis revealed that personal augmentation significantly outperformed global for experiential prompts such as “party” and “dreamy” while global augmentation was more effective for prompts with more consistent musical associations such as “mellow” and “romantic”. This suggests that the effectiveness of personalization depends on the subjective variability of the vague term.

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<sup>1</sup> [https://github.com/Atilik/NYU\\_Mtech\\_Thesis#](https://github.com/Atilik/NYU_Mtech_Thesis#)

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## 1. INTRODUCTION

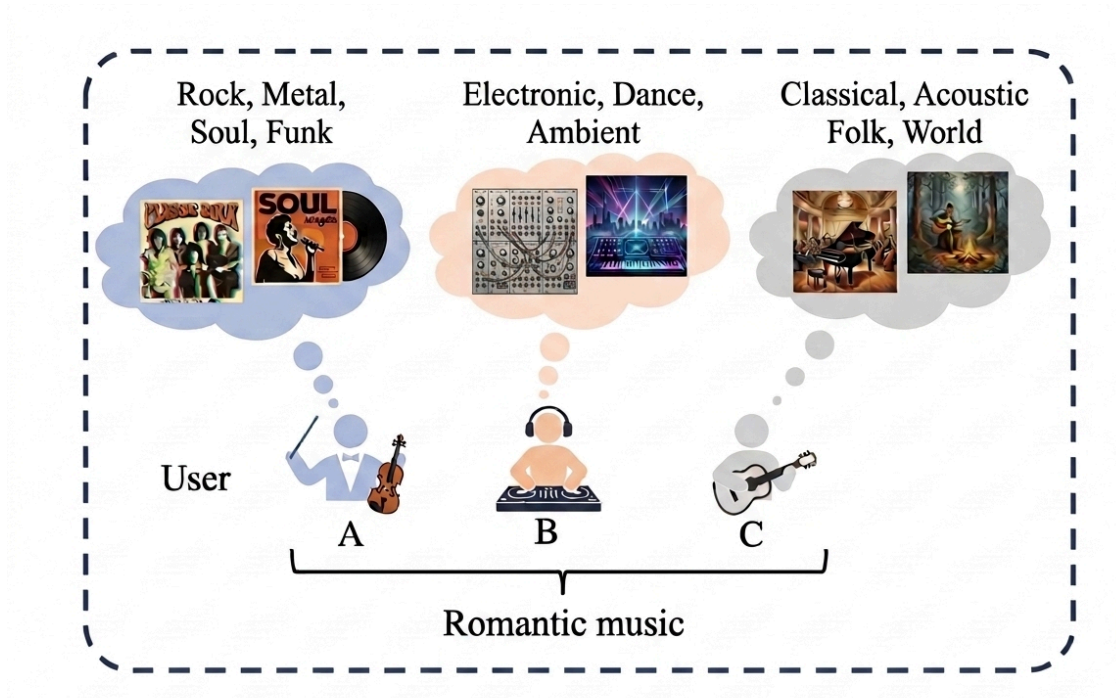
In recent years, breakthroughs in deep learning have drawn significant attention to automatic music generation, resulting in remarkable achievements. State-of-the-art models like MusicGen (Copet et al., 2023), AudioLDM 2 (Liu et al., 2024), and Stable Audio 2 (Evans et al., 2024) have notably improved automatic music generation in terms of audio quality, long-term structure, and controllability. However, despite this technical progress, they are fundamentally designed to serve general audiences, lacking automatic personalization and responsiveness to user preferences. Current models do not account for an individual user’s taste and understanding in music and often produce outputs that are far too general and underwhelming when there is not a significant prompt engineering step.

Numerous studies have explored modeling user music preferences and using them as conditioning mechanisms for the generative process. For example, Ma et al. (2022) proposed a content-based user preference modeling system that learns individual music tastes from listening histories and uses those embeddings to guide a controllable music generator, resulting in more personalized outputs. In a follow-up study, Ma et al. (2024) extended this approach by incorporating graph-based representations of user preferences and textual descriptions, improving alignment between generated music and both user taste and descriptive input. Similarly, Plitsis et al. (2023) adapted few-shot personalization techniques, originally developed for image generation, to the audio domain by fine-tuning a text-to-audio diffusion model (AudioLDM) on user-provided samples. Dai et al. (2023) approached personalization by designing a system that imitates a user’s favorite pop song using statistical modeling of structure, melody, harmony, and bass. This enabled high-quality stylistic imitations aligned with listener preferences even when limited data is available.

In their article, Ma et al. (2022, p. 1) outline four key challenges in generating music that aligns with user preferences: (1) the underlying factors that shape musical taste remain poorly understood; (2) identifying suitable features and representations for modeling preferences is difficult; (3) users often struggle to express their musical preferences,

particularly those without formal music training; and (4) incorporating user preferences into generative models as interpretable and controllable conditions is highly complex. Together, these challenges highlight why personalized automatic music generation remains an open and unsolved research problem to this day (Ma et al., 2022, p. 1).

Despite this progress in modeling user music preferences and developing controllable generative systems, to the best of the author's knowledge, no existing research has directly utilized real-world listening history metadata to augment text prompts for personalizing music generation. This thesis aims to address that gap by leveraging tag co-occurrence patterns derived from listening history metadata to personalize vague text-to-music generation prompts. When a vague prompt like "romantic" or "chill" is provided to a music generation system, the model lacks the contextual information necessary to interpret that term in a way that reflects individual semantic understanding. Figure 1 represents the different meanings that “romantic music” can have for different people.



**Figure 1.** *Conceptual illustration of personal understanding of the vague prompt “romantic music”.*

This thesis acknowledges and addresses the aforementioned challenges of personalization in automatic music generation systems.

**The central research question guiding this work is:**

Could augmenting vague text prompts with listening history metadata improve personalization in music generation?

To address this question, this thesis investigates a prompt augmentation method that appends co-occurring tags from user listening histories to vague prompts before feeding them to a music generation model. A dataset of 50 users was constructed by combining the Million Song Dataset, User Taste Profiles, and Last.fm tags. Tag co-occurrence matrices were computed both globally across all users and individually per user, resulting in three experimental conditions: unaugmented baseline, global augmentation, and personal augmentation. Music was generated using MusicGen-Large and evaluated objectively using Fréchet Audio Distance (FAD) to measure proxy personalization alignment and CLAP text-audio similarity to measure concept retention.

## 2. LITERATURE REVIEW

Automatic music generation has been explored for over sixty years through different approaches. This includes rule-based systems, probabilistic models, evolutionary algorithms, and neural networks (Ji et al., 2020). Early influences date back to the Classical period with generative techniques used by Mozart in his *Musikalisches Würfelspiel*. Starting from the 1950s, composers like John Cage and Iannis Xenakis pioneered the use of randomness and statistical models to guide composition (Herremans, Chuan, & Chew, 2017; Pritchett, 1994; Xenakis, 1992). Cope’s *Experiments in Musical Intelligence* extended this line of work by developing a system to mimic his own compositional style using algorithmic support (Cope, 1996). These experimental ideas laid the artistic groundwork for the diverse field of music generation, including melody creation and style transfer, which still continue to evolve with advances in data and deep learning (Ji et al., 2020).

Rule-based systems emerged among traditional techniques for automatic music generation. This method relies on manually encoded musical rules inspired by classical music theory (Ji et al., 2020). A classic example is Ebcioglu’s *CHORAL* that generates Bach-style chorales based on strict voice-leading rules (Ebcioglu, 1988). This method is hard, and limits generalizability and scalability to various music genres and styles (Ji et al., 2020). Another explored set of techniques is probabilistic approaches like Markov and hidden Markov models. These models offer statistical alternatives, but they often struggle with long-term coherence (Baum et al., 1970; Cope, 2000; Ji et al., 2020). Evolutionary methods such as Biles’ *GenJam* and Lavrenko and Pickens’ random field model also explored generative music generation through fitness-based search. Although these approaches tend to rely on simplified representations and abstract evaluations (Biles, 1994; Lavrenko & Pickens, 2003; Ji et al., 2020).

### ***2.1 Deep Learning Architectures Used for Music Generation***

Deep learning has increasingly become central to music generation tasks since 2012 (Ji et al., 2020, p. 6). While advancements in deep learning continue, some researchers suggest that the field is still in its early stages especially when it comes to artistic applications of

Artificial Intelligence (Ma, Wang, & Wang, 2024, p. 2). Various architectures have been explored to address the task, each architecture offering different capabilities for modeling music's timbral, textural, temporal and structural properties (Copet et al., 2023; Liu et al., 2024; Evans et al., 2024).

Recurrent Neural Networks (RNNs) were one of the first deep learning architectures used for the music generation task (Ji et al., 2020). Todd (1989) applied a basic RNN for melody generation. However, RNNs faced the vanishing gradient problem, which limited their ability to learn long-term dependencies which is crucial for musicality, making their training challenging. Later, Hochreiter and Schmidhuber (1997) addressed this issue by developing Long Short-Term Memory (LSTM) networks. Eck and Schmidhuber (2002) applied LSTM to music generation and demonstrated its ability to generate coherent blues improvisations. Later, Boulanger-Lewandowski et al. (2012) introduced an RNN-RBM hybrid network that improved musical expressiveness. Google's Magenta team later proposed Melody RNN (Waite et al., 2016) and made progress toward modeling longer-term musical sequences. Although these models were evolving, they were still struggling to capture long-term structures.

Variational Autoencoders (VAEs) offer a framework for learning representations of music (Ji et al., 2020). Roberts et al. (2018) introduced MusicVAE. Unlike RNN-based models, MusicVAE was a hierarchical model capable of creating polyphonic music sequences with long-term structure. Later, Simon et al. (2018) extended this model to support multitrack polyphonic measures and enabled operations like chord-conditioned sampling and attribute manipulation. Jia et al. (2019) also proposed a coupled latent variable model for pop music generation that synchronizes melody and accompaniment features in a shared latent space.

Generative Adversarial Networks (GANs) are less frequently applied to sequential data due to training difficulties (Ji et al., 2020). One of few examples is Yang, Chou, and Yang's (2017) MidiNet. MidiNet is a CNN-GAN-based model for conditional score generation. It allowed flexible generation of various musical styles by conditioning on

input specifications. MuseGAN, introduced by Dong et al. (2017), was one of the first GANs designed for multi-track polyphonic music generation.

The transformer architecture has become the most prominent, largely due to their effectiveness at modeling long-term dependencies (Ji et al., 2020). Huang et al.’s (2018) Music Transformer was the first model to capture long-range musical structure using relative self-attention. Donahue et al. (2019) extended this idea to multi-instrumental music and introduced pre-training via transfer learning. Other state-of-the-art models such as MusicGen (Copet et al., 2023), MusicLM (Agostinelli et al., 2023), and JEN-1 (Li et al., 2023) use transformer architecture as a core due to its ability to learn long-term sequential information, which is crucial for music generation.

Diffusion models such as Stable Audio Open (Evans et al., 2024) represent another direction in deep learning for music generation. These models iteratively denoise latent representations to generate complex structures (Evans et al., 2024, p. 1). AudioLDM 2 (Liu et al., 2024) and Stable Audio Open (Evans et al., 2024) use diffusion architecture as a core for their ability to generate high-quality audio in the audio domain. However, they require high computational power and resources to train, infer, and fine-tune (Evans et al., 2024).

These state-of-the-art models, MusicGen (Copet et al., 2023), MusicLM (Agostinelli et al., 2023), YuE (Yu et al., 2025), JEN-1 (Li et al., 2023), AudioLDM (Liu et al., 2024, pp. 1–2), and Stable Audio Open (Evans et al., 2024), combine both architectures to leverage the strengths of each one.

## ***2.2 State-of-the-Art Automatic Text-to-Audio Generation Models***

MusicGen is a transformer-based model introduced by Meta AI (Copet et al., 2023). Copet et al. (2023, p. 4) state that MusicGen supports both text and audio conditioning and is trained exclusively on music data for the purpose of music generation. The model is fully open source and designed to be compatible with fine-tuning workflows (Copet et al., 2023, p. 1). Compared to diffusion-based models, MusicGen uses an autoregressive

transformer decoder to generate discrete audio tokens and eliminates the need for a denoising process (Copet et al., 2023, p. 2). However, MusicGen is not able to model long-term musical structure effectively, has no support for multi-track or instrument-level control, has fixed output durations, has limited prompt control over specific musical sections, and has a potential quality loss due to its EnCodec tokenization process (Copet et al., 2023, p. 10).

AudioLDM 2 is another state-of-the-art latent diffusion-based audio generation model that introduces a modular architecture involving three components; a Language of Audio (LOA) Translator, a Self-supervised LOA Generator, and a Universal Vocoder (Liu et al., 2024, pp. 1–3). In their article, Liu et al. (2024, pp. 3-7) explains that the model supports multi-modal conditioning using audio, text, or transcription inputs, which are projected into a shared representation space using pretrained modules such as CLAP, FLAN-T5, and a phoneme encoder. These are processed by a GPT-2 model to generate a semantic representation referred to as the "Language of Audio" (LOA). LOA is then fed into a latent diffusion model with a transformer-based UNet architecture for iterative denoising in latent space (Liu et al., 2024, pp. 3–7). Finally, the generated latent is decoded back into waveform audio using a VAE-based universal vocoder (Liu et al., 2024, pp. 3–7). Because of its flexible multi-modal conditioning pipeline and learned audio representation, AudioLDM 2 enables diverse tasks such as text-to-audio, text-to-music, and text-to-speech (Liu et al., 2024, p. 1). However, the model is not trained specifically on musical data or designed especially for text-to-music generation (Liu et al., 2024, pp. 8-11). Also, the diffusion-based generation process requires substantial computational resources for both inference and fine-tuning (Liu et al., 2024, p 2).

Stable Audio Open (Evans et al., 2024) is another open-source model that supports both text and audio conditioning and is designed to be fine-tuneable. At its core, it uses a latent diffusion architecture and generates audio by iteratively denoising latent representations (Evans et al., 2024, p. 1). In the article, Evans et al. (2024, pp. 1–2) state that the model architecture consists of a VAE-based autoencoder for audio compression, a T5-based encoder for text conditioning, and a transformer-based diffusion model (DiT) operating in

the latent space. It supports variable-length stereo audio generation at 44.1 kHz and is released with full weights and code that promotes transparency, reproducibility, and accessibility for the research and creative communities (Evans et al., 2024, p. 1). In their paper, Evans et al. (2024, p. 2) state that Stable Audio Open is trained on a dataset composed entirely of Creative Commons-licensed audio from sources such as Freesound (Freesound, n.d.) and FMA (Defferrard et al., 2016), which covers a wide range of sound effects, foley sounds, and music. Because Stable Audio Open is trained on both sound effects and music, it is suitable for general audio synthesis tasks like sound effects generation and ambient texture generation; but music generation is less competitive, and the model is said to be unable to generate intelligible speech (Evans et al., 2024, p. 4).

YuE is a relatively new text-to-audio foundation model specifically designed for long-form lyrics-to-song generation, built on the LLaMA 2 architecture (Touvron et al., 2023) and capable of producing multi-minute music with vocal, accompaniment, and lyrical alignment (Yu et al., 2025, p. 6). It introduces novel techniques such as track-decoupled next-token prediction, structural progressive conditioning, and redesigned music in-context learning, which enable high-quality musical structure and style control across multiple languages and genres (Yu et al., 2025, p. 7). In their results, Yu et al. (2025, pp. 16–21) report that the YuE model significantly advances long-form lyrics-to-song generation by addressing key challenges such as signal complexity, long-range dependencies, and data scarcity compared to other state-of-the-art models. However, YuE's semantic-fused tokenization method limits its vocal/accompaniment acoustic detail, and the model requires large-scale computational resources for effective training and inference (Yu et al., 2025, pp. 16–21).

### ***2.3 Conditioning Approaches for Music Generation***

In their article, Li et al. (2023) state that conditioning generative models typically involves adopting either autoregressive (AR) or non-autoregressive (NAR) paradigms. The models discussed above, such as MusicLM (Agostinelli et al., 2023) and MusicGen (Copet et al., 2023), employ autoregressive conditioning, where outputs are generated sequentially. In contrast, AudioLDM 2 (Liu et al., 2024) uses a non-autoregressive

approach and generates audio in parallel through diffusion. Recent music generation models increasingly rely on conditioning strategies to control and guide the output (Copet et al., 2023, p. 2; Evans et al., 2024, p. 1; Liu et al., 2024, p. 1), resulting in three primary forms of conditioning: text-based, audio-based, and symbolic (MIDI or score) conditioning (Ji et al., 2022).

Text-based conditioning enables models to generate music from natural language descriptions. AudioLDM (Huang et al., 2023) uses a pretrained CLAP text encoder to embed user prompts and a latent diffusion model to synthesize audio conditioned on these embeddings. MusicLM (Agostinelli et al., 2023) introduces a two-stage autoregressive transformer conditioned on discrete semantic tokens, where text prompts are embedded via a dual-encoder model (MuLan). MusicGen (Copet et al., 2023) uses a single-stage transformer model that processes all audio codec tokens at once and injects text prompts via cross-attention. In all three models mentioned above, the key technique used to ensure that the generated audio aligns with the given text prompt is the application of classifier-free guidance or prompt-based embedding injection, which enables descriptive control over instrumentation, mood, or genre.

Another approach is audio-based conditioning, such as in MusicGen (Copet et al., 2023), which allows generation models to incorporate specific content or style from a user-provided audio input. MusicLM (Agostinelli et al., 2023) and MusicGen (Copet et al., 2023) both offer melody conditioning, where an input melody, either sung or extracted from audio, is encoded and passed into the generation model to guide pitch and rhythm. In MusicGen (Copet et al., 2023), melody tokens are concatenated into the transformer input. Another model called SingSong (Donahue et al., 2023) uses vocal tracks as input to generate accompaniment by encoding the vocals into a discrete representation and using it as context for a transformer-based generator.

A third method is symbolic conditioning. This method uses structured musical representations like MIDI, chords, or beat patterns to guide generation (Ji et al., 2020). MusiConGen (Lan et al., 2024) and MelodyDiffusion (Li & Sung, 2023) use chord

sequences and rhythmic cues to constrain melodic generation by injecting tokenized chord or rhythm vectors into the input sequence or model attention layers.

#### ***2.4 User Music Preference Modeling for Music Generation***

Passino et al. (2021, p. 1) state that over thirty years of research and practice have shown that understanding user preferences and tailoring experiences around them significantly improves user engagement and satisfaction.

Two main methods have been widely used for modeling User Music Preferences (UMP) in music recommendation: collaborative filtering (CF) and content-based filtering (CBF) (Ma, Wang, & Wang, 2024, p. 2). CF algorithms recommend items based on user profiles and listening history (Natarajan et al., 2020; Hu, Koren, & Volinsky, 2008). However, Ma et al. (2024, p. 2) point out that CF doesn't capture why users prefer what they do, making it less effective for music generation.

On the other hand, CBF creates dense embeddings from user interactions and uses them to predict future preferences (Kant & Bharadwaj, 2012). But again, as Ma et al. (2024, p. 2) mention, these embeddings struggle to support music generation tasks because it's unclear which musical traits they encode, making integration difficult. Hybrid approaches combining both methods have also been explored (Oramas, Nieto, Sordo, & Serra, 2017), and deep learning models are now being used to better capture UMPs (Ma et al., 2024, p. 2).

In their study, Content-based User Preference Modeling in Music Generation, Ma et al. (2022) introduce a hierarchical convolutional recurrent neural network with self-attention (CRNN-SA) to extract user preferences and map them into a structured embedding space. In this space, common preferences are near the center while rare ones lie further out. They propose an embedding distance model that compares a user's profile to a seed song across four musical features: chord progression, melody statistics, rhythmic patterns, and melodic contour (Ma et al., 2022, p. 2). This distance guides the generation system to

create music that is closer to the user's taste and acts as a bridge between preference modeling and controllable music generation (Ma et al., 2022, p. 2).

In a newer study, Ma, Wang, and Wang (2024a) build on their 2022 work by shifting from a distance-based embedding model to a graph-learning-based approach for user music preference modeling. In contrast to their previous research, this 2024 paper constructs user-item graphs from listening histories, representing users and songs as nodes connected by interaction edges (Ma et al., 2024). These graphs are used to learn user embeddings through self-supervised methods (Ma et al., 2024). Additionally, the model incorporates textual queries by converting them into virtual nodes and enables alignment between user intent and music generation (Ma et al., 2024). Ma et al. (2024) then used these learned embeddings to guide a symbolic music generation system. Ma et al. (2024, p. 12) state that their experimental objective and subjective evaluation results show that the proposed user music preference model is effective in guiding the generation of symbolic music from scratch that meets personal music preferences and matches textual queries.

Plitsis et al. (2023) adapt few-shot personalization methods from computer vision to music by fine-tuning text-to-audio diffusion models like AudioLDM using user-provided samples. They use techniques like Textual Inversion and DreamBooth to inject user-specific musical traits into the model through a placeholder token. The evaluation includes embedding-based similarity, musical features like rhythm and loudness, and a user study (Plitsis et al., 2023).

Dai et al. (2023) present a symbolic music generation system that creates personalized outputs by imitating the structure and musical style of a user-selected seed song. The model extracts melody, chords, bass, and formal structure, and uses statistical learning to reconstruct a full-length song that resembles the original in style (Dai et al., 2023).

Chen et al. (2024) introduce a diffusion-based symbolic music generation model that combines user embeddings and textual prompts to generate personalized music. The

system learns a joint embedding space between users and music, calculates similarity using cross-modal embeddings, and fine-tunes the generation process through a learned style transfer network (Chen et al., 2024).

### ***2.5 Tag Co-occurrence and Music Metadata***

Tag co-occurrence has been used to understand relationships between musical concepts. When two tags frequently appear on the same songs, they likely share a meaningful connection. Pointwise Mutual Information (PMI) is a standard method for identifying such associations and measures whether two items appear together more often than expected by chance (Church & Hanks, 1990). PMI has been widely adopted in natural language processing and distributional semantics (Levy & Goldberg, 2014).

Platforms like Last.fm allow users to tag songs with descriptive labels such as genre, mood, and instrumentation. In the Music Information Retrieval (MIR) community, Lamere (2008) explored using these social tags for music information retrieval, noting both their richness and their noise.

The Million Song Dataset (Bertin-Mahieux et al., 2011) is one of the largest publicly available datasets for music research, containing metadata and audio features for one million songs. The Taste Profile subset extends it with user listening histories. Combined with Last.fm tags, this creates a resource for studying the relationship between listening behavior and musical semantics.

A key challenge in text-to-music generation is interpreting vague prompts. Zang and Zhang (2024) identify this as the "interpretation gap," where a single vague prompt can correspond to many valid musical outputs. This thesis uses tag co-occurrence patterns from listening histories to address this gap.

### ***2.6 Evaluation Methods for Music Generation***

In their survey, Ji et al. (2020, p. 65) state that there is no single unified evaluation criterion for generative algorithms such as those used in music generation. Ji et al. (2020,

p. 65) also note that there are two main categories for evaluating such systems: objective and subjective. Objective evaluations include model metrics, which compare the generated data to the original data, as well as music metrics and descriptive statistics. Subjective evaluations involve listening tests or visual analysis (Ji et al., 2020, pp. 65–76).

The Fréchet Audio Distance (FAD) is a method that compares the statistical distribution of generated audio embeddings to that of real music references (Kilgour, Zuluaga, Roblek, & Sharifi, 2019). FAD primarily measures the audio quality of the output, and it has been used alongside adherence metrics to ensure that the model has not sacrificed quality for adherence (Gonzales & Rudzicz, 2024).

A common approach for evaluating prompt adherence is to use the CLAP score, which relies on a Contrastive Language-Audio Pretraining model to embed both the text description and audio, then compute their cosine similarity (Elizalde, Deshmukh, Al Ismail, & Wang, 2022).

Another objective evaluation method was introduced with Google’s MusicLM, using their MuLan text-audio embedding model (Agostinelli et al., 2023). In their paper, Agostinelli et al. (2023) computed MuLan embeddings for MusicCaps dataset descriptions and the corresponding generated audio, then defined an alignment score as the average cosine similarity, referred to as MuLan Cycle Consistency.

Parker et al. (2024) proposed the Music Information Retrieval Descriptor Distance (MIRDD), which computes the Kullback–Leibler divergence between distributions of musical features, such as pitch, rhythm, and structure, in the conditioning audio versus the generated audio.

Grachten (2024) introduces a model-agnostic framework for evaluating prompt adherence in audio generation systems using embedding-based distance metrics. The proposed method relies on pre-trained audio and text encoders such as CLAP, VGGish, or

OpenL3 to extract high-dimensional representations of both prompts and generated outputs (Grachten, 2024). These embeddings are then compared using metrics like Fréchet Audio Distance (FAD) and Maximum Mean Discrepancy (MMD) to quantify similarity. The evaluation pipeline supports different strategies for combining prompt and output, including mixing, summing, and concatenation, and finds that early fusion (mixing the audio before embedding) produces the most consistent results (Grachten, 2024). This study also introduces a relative proximity score that evaluates whether matched prompt-output pairs are closer than mismatched ones and offers a robust way to assess prompt alignment across different conditions.

Subjective evaluation methods are also commonly used, such as the Mean Opinion Score method (Hong et al., 2024) or A/B comparisons (Agostinelli et al., 2023), to complement the results from the objective methods discussed above.

## ***2.7 Conclusion***

In this literature review, the current state-of-the-art deep learning-based text-to-music generation models are explored, showing that transformer and diffusion-based hybrid architectures dominate the field. Different conditioning methods are presented; text, audio, and symbolic. Recent works on the personalization of music generation models are examined, showing that user listening history and its use as a conditioning mechanism in deep learning-based music generation is potentially effective. Several evaluation methods are also discussed, revealing that both objective metrics like CLAP and FAD, and subjective tests like MOS, are commonly used. This literature review highlights a gap in automatically generating music customized to an actual user, especially leveraging user listening histories to personalize text-to-music generation prompts.

### 3. METHOD

This chapter explains the experimental design used to develop and evaluate a prompt augmentation method for personalizing vague text prompts for music generation. Specifically, this study explores whether Pointwise Mutual Information (PMI)-weighted tag co-occurrence patterns derived from semantic tags associated with a user’s listening history metadata can augment ambiguous text prompts and steer MusicGen outputs closer to user-specific reference audio distributions associated with the same target tag.

To evaluate this approach, this study adopts and compares three conditions: **(1)** an unaugmented vague-prompt baseline, **(2)** a global augmentation condition, and **(3)** a personal augmentation condition. In the global condition, PMI-weighted tag co-occurrence patterns are computed across the full dataset of listening histories. In the personal augmentation condition, PMI-weighted tag co-occurrence patterns are computed separately for each individual user. Comparing these conditions makes it possible to examine whether user-specific augmentation provides an additional benefit over both the original vague prompt and a non-personalized (global) augmented prompt in the context of automatic personalization of music generation.

This section details the research questions, datasets used, experimental design, music generation procedure, evaluation framework, and data analysis methods.

#### *3.1 Research Questions*

This study addresses the following main research question and four sub-research questions:

**Main RQ:** Does augmenting vague text prompts with listening-history-derived metadata improve personalization in text-to-music generation?

- **Sub-RQ1:** Does personalized augmentation outperform the unaugmented baseline at the user level?
- **Sub-RQ2:** Does personalized augmentation outperform global augmentation at the user level?

- **Sub-RQ3:** Do augmented prompts preserve alignment with the original vague concept?
- **Sub-RQ4:** Which vague prompts benefit most from personalized augmentation?

### ***3.2 Datasets Used***

To address these research questions in a quantitative manner, a dataset with user listening histories, corresponding song metadata, and descriptive tags for each song was required. To obtain this dataset, several open datasets and subsets were combined.

The main dataset used in this study is the Million Song Dataset<sup>2</sup> (MSD; Bertin-Mahieux et al., 2011). The MSD provides song-level metadata such as artist names, album names, track titles, duration, and pre-computed audio features, including key, mode, and tempo. In this study, the MSD served as the main metadata source used to identify songs and link them across the other datasets.

The Taste Profile subset<sup>3</sup>, an extension of the MSD (McFee et al., 2012), was used to include user listening activity. This subset contains triplets of user ID, song ID, and play count. The song ID was used to link this subset to the MSD, which carried the same ID for songs.

Lastly, the Last.fm dataset<sup>4</sup> (Bertin-Mahieux et al., 2011; Million Song Dataset, n.d.) was combined to provide human-annotated tags for each song. This dataset contains user-generated tags per song covering mood (e.g., "romantic", "melancholic"), instrumentation (e.g., "piano", "guitar"), texture (e.g., "acoustic", "ambient"), and feel (e.g., "dreamy", "groovy"). These semantic tags also come with a weight value between 0 and 100 indicating how strongly users associated that tag with the song. This dataset was the primary source for building the PMI-weighted co-occurrence matrices.

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<sup>2</sup> <http://millionsongdataset.com/>

<sup>3</sup> <http://millionsongdataset.com/tasteprofile/>

<sup>4</sup> <http://millionsongdataset.com/lastfm/>

From the final combined dataset, 50 users were selected based on the number of songs in their listening histories and the availability of Last.fm tags for those songs. A minimum weight threshold of 5 was applied to filter out noisy tags in the dataset. For each selected user, 500 songs were retained by selecting the songs with the highest number of Last.fm tags after the filter. This resulted in a final dataset of 25,000 rows, covering 17,003 unique songs. Every song in the final dataset had at least one Last.fm tag after filtering, resulting in 100% tag coverage. The final dataset was then split into two parts: (1) **Split A** contained 80% of each user’s songs and was used as the development set for creating the augmented prompts, and (2) **Split B** contained the remaining 20% and was kept separate as the evaluation set for building the reference audio sets. Table 1 summarizes the datasets used in this study and their roles in the experimental pipeline.

| Dataset              | Content                                            | Role                                             |
|----------------------|----------------------------------------------------|--------------------------------------------------|
| Million Song Dataset | Song ID, artist, title, duration, key, mode, tempo | Backbone, song metadata, and high level features |
| Taste Profile Subset | User ID, song ID, play count                       | Creating listening histories                     |
| The Last.fm Dataset  | User-generated tags with weights (0-100)           | Source for the tag co-occurrences                |

**Table 1.** *Datasets used, their content, and their roles in the experimental design.*

### 3.3 Experimental Design

From the human-annotated tags in the Last.fm dataset, several vague prompts that could be augmented into more meaningful text prompts were identified. In this study, a “vague prompt” is defined as a text prompt that describes subjective musical qualities such as mood, atmosphere, or activity, rather than concrete musical attributes such as key, instruments, or tempo. Since the same word can have different musical meanings for different users, these vague terms are well suited for testing whether PMI-based tag co-occurrence patterns can support personalized prompt augmentation.

The vague prompts selected for the experiment were “chill”, “mellow”, “melancholy”, “dark”, “romantic”, “happy”, “energetic”, “dreamy”, “party”, and “catchy.” These 10 prompts were selected based on tag frequency, with each term required to appear on at least 50 songs with a tag weight of 5 or higher across all 50 users. This threshold was used to ensure sufficient data for constructing co-occurrence matrices and held-out reference sets for evaluation (e.g., "chill" appeared on 5,014 songs across all 50 users).

Human-annotated Last.fm tags are unsupervised and contain noise. To mitigate this, a multi-step filtering pipeline was applied in addition to the tag weight threshold of 5. The original vague prompt and its morphological variants (e.g., "chills" for "chill") were removed to prevent self-augmentation. Artist and band names (4,479 entries from the MSD) were excluded. Opinion and bookmark tags such as "favorite," "awesome," and "seen live" were removed. Demographic and nationality tags (e.g., "british," "female vocalists"), named entities (e.g., organizations), and tags containing special characters or exceeding 25 characters were also discarded. Lastly, a junk detection function removed sentence-like tags (four or more words), tags starting with personal prefixes ("my," "i"), and tags containing numeric characters. In total, 20,841 tags were excluded from the augmentation vocabulary.

To address the ambiguity of these tags, co-occurrence matrices were constructed from the dataset to expand ambiguous tags with additional co-occurring tags. A co-occurrence matrix counts how often pairs of tags appear together on the same song and normalizes these counts into conditional probabilities. For example, if 35% of songs tagged “romantic” are also tagged “harp,” then:

$$P(harp \mid romantic) = 0.35$$

However, empirical evaluations showed that this approach led to augmenting mostly globally frequent tags in the Last.fm dataset, such as “rock,” “indie,” and “alternative,” which appeared across many songs. As a result, the augmented prompts often reflected

general popularity patterns in the dataset rather than distinctive associations with a particular vague term.

To reduce the bias of globally frequent tags appearing in every augmentation, Pointwise Mutual Information (PMI) was used. PMI measures whether two tags co-occur more often than would be expected by chance (Church & Hanks, 1990). For each vague prompt  $v$  and candidate tag  $t$ ,  $P(v)$  and  $P(t)$  denote the probability of each tag appearing in the relevant song set, and  $P(v, t)$  denotes the probability of both tags appearing on the same song. To retain only positively associated tags, negative PMI values were set to zero. This method is called Positive PMI (PPMI; Levy & Goldberg, 2014). Lastly, PPMI scores were multiplied by a logarithmic co-occurrence count term to ensure that selected tags were both distinctive and sufficiently attested in the data.

The formula is as follows:

$$PPMI(v, t) \equiv \max(\log_2 \frac{P(v,t)}{P(v)P(t)}, 0)$$

$$Score(v, t) \equiv PPMI(v, t) \cdot \log_2(count(v, t) + 1)$$

where  $v$  is the vague prompt,  $t$  is a candidate tag, and  $count(v, t)$  is the number of songs containing both tags. In the global condition, these probabilities were computed across all users' listening histories in the development set. In the personal condition, they were computed separately within each user's listening history in the development set.

For each vague prompt, candidate tags were ranked using this frequency-weighted PPMI score. The top-ranked tags were then appended to the original vague prompt as a comma-separated list, up to 10 tags in total. No natural language template or additional MSD audio-feature descriptors were added to the prompt.

The proposed method was evaluated under three conditions:

1. **Unaugmented:** The vague prompt was used unchanged. For example, the prompt “chill” was provided directly to MusicGen without additional tags. This condition served as the baseline.
2. **Global augmentation:** PMI-weighted co-occurrence scores were computed across all users. For example, the prompt “dark” was expanded as "dark, gothic, darkwave, gothic rock, post punk, industrial, goth rock, trip hop, haunting, industrial metal, industrial rock". This condition was used to test whether dataset-level tag associations improve vague prompts without user-specific personalization.
3. **Personal augmentation:** PMI-weighted co-occurrence scores were computed separately for each user using that user’s listening history. This produced a different augmented prompt for each user. For example, for one user, “dark” was expanded as "dark, synthpop, indietronica, experimental, electropop, 2000s, instrumental, ambient, electronica, chillout, alternative". This condition was used to test whether user-specific tag associations provide an additional benefit over global augmentation.

### ***3.4 Music Generation Procedure***

The open-source music generation model MusicGen-Large with 3.3 billion parameters (Copet et al., 2023) was used to generate music for each of the 10 vague prompts identified across three prompt conditions: unaugmented, global augmentation, and personal augmentation. Each prompt-condition pair was used to generate 100 clips with deterministic seeds to account for the stochastic nature of the generation process.

The default recommended generation parameters from the MusicGen paper (Copet et al., 2023) were used for generation: a duration of 10 seconds, top- $k$  sampling with  $k = 250$ , a classifier-free guidance coefficient of 3.0, and sampling enabled with a temperature of 1.0. As confirmed by the AudioCraft codebase, all dropout mechanisms (word dropout, classifier-free guidance dropout, and attribute dropout) are automatically disabled during inference, meaning that every tag in augmented prompts is fully processed by the model’s T5 text encoder (Raffel et al., 2023).

In order to reduce computational time and complexity, 10 representative users were selected for the personal condition out of 50 based on having the most non-empty personal prompts across all 10 vague tags. In total, 1,000 unaugmented clips, 1,000 global augmented clips, and 10,000 personal augmented clips were generated, resulting in 12,000 ten-second music clips. Generation was completed on NYU HPC Torch, using L40 GPUs in an array job that took approximately 7 hours.

### ***3.5 Evaluation Framework***

To evaluate the proposed method, this study uses objective audio-embedding-based metrics to compare the unaugmented, global augmentation, and personal augmentation conditions. The primary evaluation metric is Fréchet Audio Distance (FAD), which measures distributional similarity between generated audio and a held-out reference set of audio (Kilgour et al., 2019). In this study, FAD is used to evaluate whether augmented prompts produce MusicGen outputs that are closer to user-specific reference audio collected from each user’s held-out listening history.

Additionally, Contrastive Language-Audio Pretraining (CLAP) text-audio similarity (Elizalde et al., 2022) is used to assess whether augmented prompts preserve alignment with the original vague concept after augmentation. For each generated clip, the audio embedding is compared with the text embedding of the original vague prompt rather than the full augmented prompt. This makes it possible to test whether augmentation causes generated audio to shift away from the original target concept in the CLAP embedding space.

In addition to these main analyses, CLAP audio-audio similarity and a user-ranking analysis are reported as secondary exploratory evaluations. CLAP audio-audio similarity measures the average similarity between generated audio and held-out reference audio in the CLAP embedding space. The user-ranking analysis evaluates whether generated clips are more similar to the correct user’s reference distribution than to other users’ reference

distributions for the same prompt. These analyses provide additional context but are treated as exploratory rather than primary evidence.

### ***3.5.1 Reference Set Construction***

To avoid data leakage and circular evaluation, the final dataset was split into two parts. The curated dataset contained 50 users, each with 500 songs, resulting in 25,000 user-song rows. Split B, the evaluation set, retained 20% of the whole dataset. More specifically, 100 songs from each user were held out to create this evaluation set resulting in a total number of 5000 songs. This split B was not used for prompt augmentation and only used in the construction of the reference set. The split B also took into consideration that random selection could lead to various vague prompts that did not appear in the evaluation split. To avoid this problem, at least one reference song for each vague prompt was made sure to be kept in this set

The remaining 80% of the data (20000 songs) was used as the development set, split A. This split was used to compute PMI-weighted tag co-occurrences and create augmented prompts.

Reference songs from Split B were downloaded using a Python script that searched YouTube using each song’s artist name, title, and duration. Downloaded audio files were converted to mono and resampled to 32,000 Hz to match the format of the generated MusicGen clips. Reference tracks were divided into 10-second non-overlapping chunks before embedding to match the 10-second duration of the generated clips.

In total, 1,825 reference songs were successfully downloaded. For FAD computation, only user-prompt pairs with at least three reference songs were retained, resulting in 83 valid user-prompt pairs out of 100.

### ***3.5.2 Fréchet Audio Distance (FAD)***

The Fréchet Audio Distance (Kilgour et al., 2019) is a reference-based metric that measures the distributional similarity between two sets of audio. In the context of this

thesis, these two sets are the generated music and a held-out set of reference music. Unlike per-sample metrics, FAD evaluates the statistical properties of entire sets of audio clips.

FAD operates by extracting embeddings from both the generated audio and the reference audio using a pretrained audio encoder. In this study, CLAP audio embeddings (Elizalde et al., 2022) were used instead of the traditional VGGish embeddings. CLAP was selected because it provides audio representations that may better capture musically relevant information for this evaluation.

Multivariate Gaussian distributions are fitted to both the generated and reference embedding sets, characterized by their mean vectors ( $\mu$ ) and covariance matrices ( $\Sigma$ ). The Fréchet distance between these two distributions is then computed. A lower FAD score indicates that the generated audio is distributionally closer to the reference set. The formula is as follows:

$$FAD = \left\| \mu_{gen} - \mu_{ref} \right\|^2 + Tr(\Sigma_{gen} + \Sigma_{ref} - 2\sqrt{\Sigma_{gen}\Sigma_{ref}})$$

where  $\mu$  and  $\Sigma$  denote the mean and covariance of the generated (gen) and reference (ref) embedding distributions, and Tr denotes the matrix trace.

In this experimental design, for each user and each vague prompt, the reference set consists of songs from that user's held-out split (Split B) that are tagged with the associated term. Reference songs were downloaded from YouTube at 32,000 Hz mono and used at full length. For fair comparison with the 10-second generated clips, reference audio was chunked into 10-second non-overlapping segments before embedding extraction. FAD was computed separately for each user, prompt, and condition against this shared per-user reference set. This allows measurement of whether augmented prompts produce music that is distributionally closer to what each individual user listens to for a given vague term.

### ***3.5.3 CLAP Text-Audio Similarity***

CLAP (Elizalde et al., 2022) is a pre-trained model that uses contrastive learning to align audio and text in a shared embedding space. Trained on large-scale audio-text pairs, it is commonly used to compute cosine similarity as a proxy for how well an audio clip matches a given text description.

In this thesis, CLAP text-audio similarity is adopted as a secondary metric to evaluate whether augmented prompts preserve alignment with the original vague concept. Adding tags to a prompt may move the generated audio closer to a user's listening history while simultaneously shifting it away from the intended meaning of the original prompt.

For this analysis, the original vague prompt text, such as “romantic” or “chill,” was embedded using the CLAP text encoder. Each generated audio clip was embedded using the CLAP audio encoder. Cosine similarity was then computed between the text embedding of the original vague prompt and the audio embedding of each generated clip. A higher similarity score indicates stronger alignment with the original vague concept.

The same original vague prompt was used as the text target for all three conditions. For example, clips generated from “romantic,” from the global augmented version of “romantic,” and from the personal augmented version of “romantic” were all compared against the CLAP text embedding of “romantic.” This allowed the analysis to test whether augmented prompts preserved alignment with the original vague concept.

### ***3.5.4 Secondary Metrics***

Two supplementary analyses were also completed. CLAP audio-audio similarity was used to measure the cosine similarity between CLAP audio embeddings of generated clips and reference audio from each user's held-out set. This provides a direct audio-level comparison without relying on text, similar to FAD.

Additionally, inspired by the retrieval evaluation approach used in the MSD challenge (McFee et al., 2012), a user ranking analysis was conducted. For each user, prompt, and

condition, the mean CLAP embedding of the generated clips was compared against the mean CLAP embedding of each user's reference set for that prompt. Users were ranked by cosine similarity, and the rank of the correct target user was recorded. Hit rate at top-1 (HR@1), hit rate at top-3 (HR@3), and mean reciprocal rank (MRR) are reported.

### ***3.5.5 Statistical Analysis***

Statistical comparisons were conducted between matched user-prompt pairs. For FAD, each user-prompt pair produced one score per condition. Paired t-tests were then used to compare personal augmentation against both the unaugmented baseline and the global augmentation condition across these matched pairs.

For CLAP text–audio similarity, per-prompt mean scores were computed for each condition, and paired t-tests were conducted across the 10 prompts.

Paired t-tests were used to examine whether differences between conditions were statistically significant. Effect sizes were reported using Cohen's d. In addition to overall condition-level comparisons, per-prompt analyses were completed to identify which vague prompt benefited most from personal augmentation and which prompts showed stronger performance under global augmentation.

All statistical tests were implemented in Python using SciPy and can be found in the evaluation notebook.

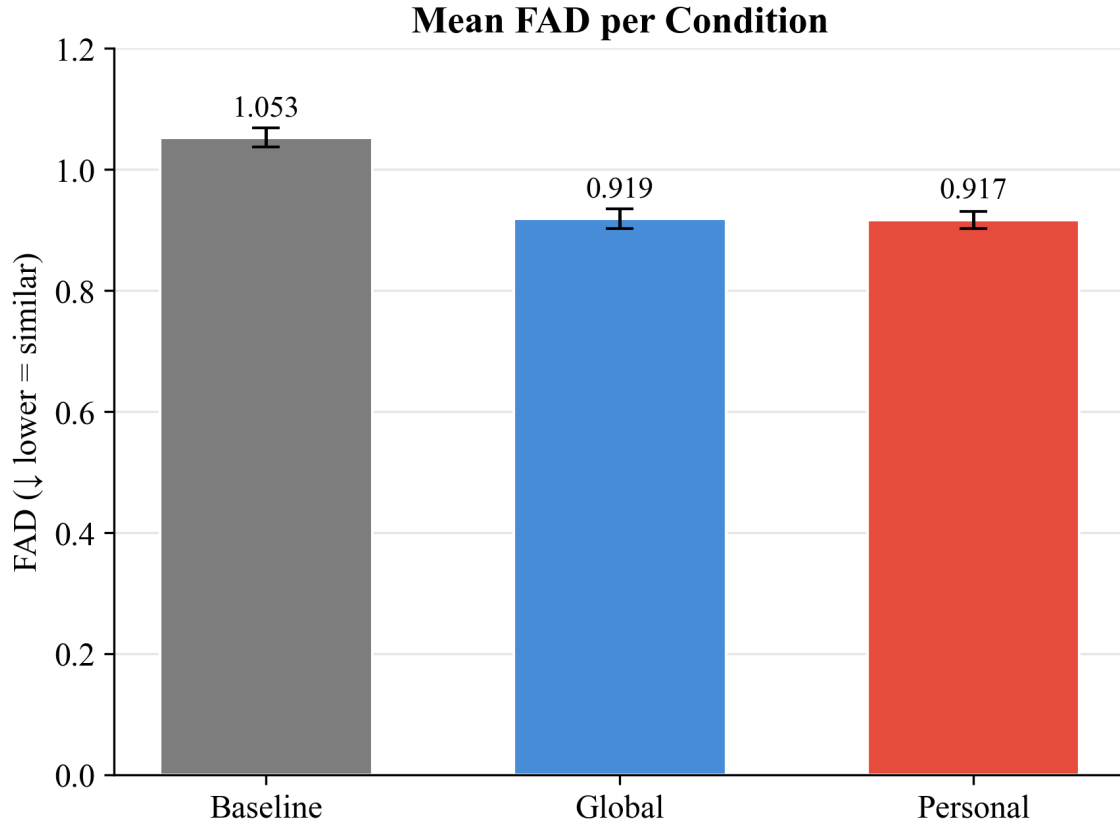
## 4. ANALYSIS

This chapter presents the results and analysis of the conducted experiment, and discusses their implications in relation to the research questions.

### 4.1 Quantitative Results

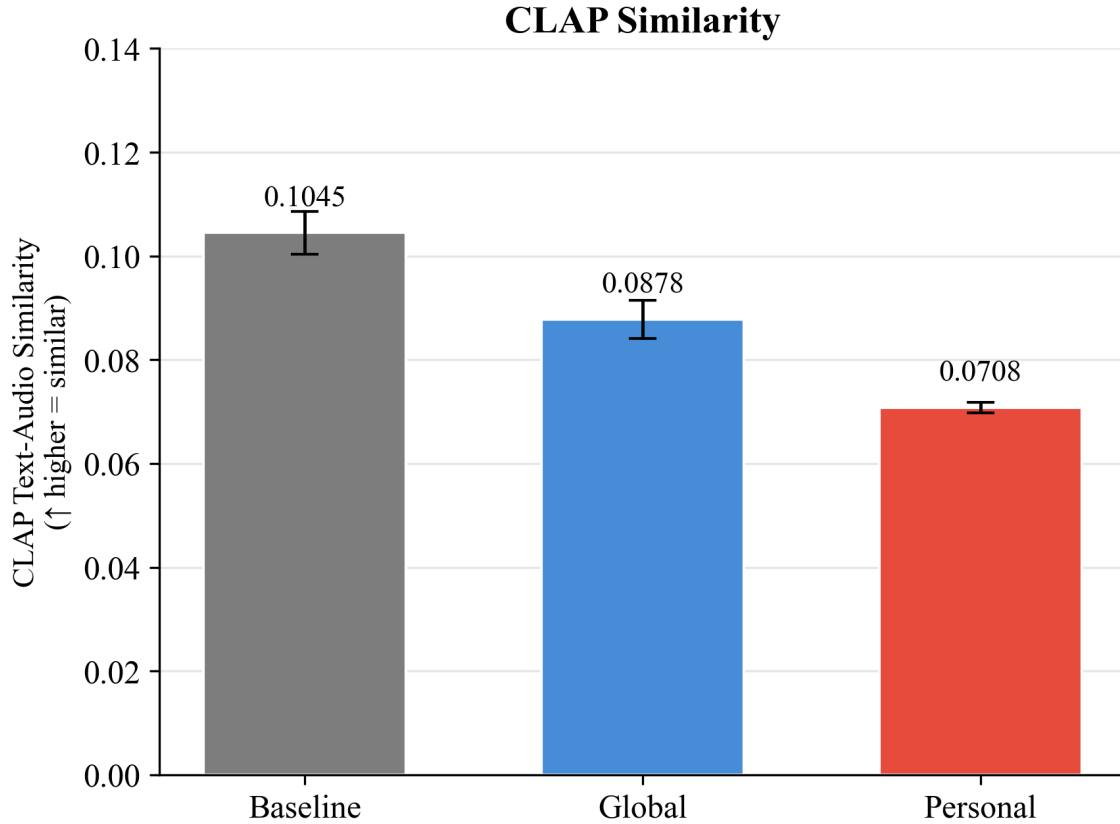
#### 4.1.1 Mean Comparison Across Conditions

Figure 2 presents the mean FAD calculations across the three conditions. Lower FAD indicates that the generated music was distributionally closer to the user's held-out reference music. The unaugmented baseline produced the highest mean FAD at 1.05. Both global (0.919) and personal (0.917) augmentation achieved substantially lower scores. The difference between personal and baseline was statistically significant ( $p < 0.0001$ ,  $d = 0.84$ ), as was global versus baseline ( $p < 0.0001$ ,  $d = 0.95$ ). However, the difference between personal and global was not significant ( $p = 0.89$ ,  $d = 0.01$ ).



**Figure 2.** Mean Fréchet Audio Distance (FAD) score across the three conditions.

Figure 3 presents the mean CLAP text-audio similarity scores across the same three conditions. Higher values indicate higher similarity with the original vague prompt. The unaugmented condition scored 0.105, global 0.088, and personal 0.071. Although augmented conditions scored lower, none of the differences reached statistical significance (all  $p > 0.07$ ).

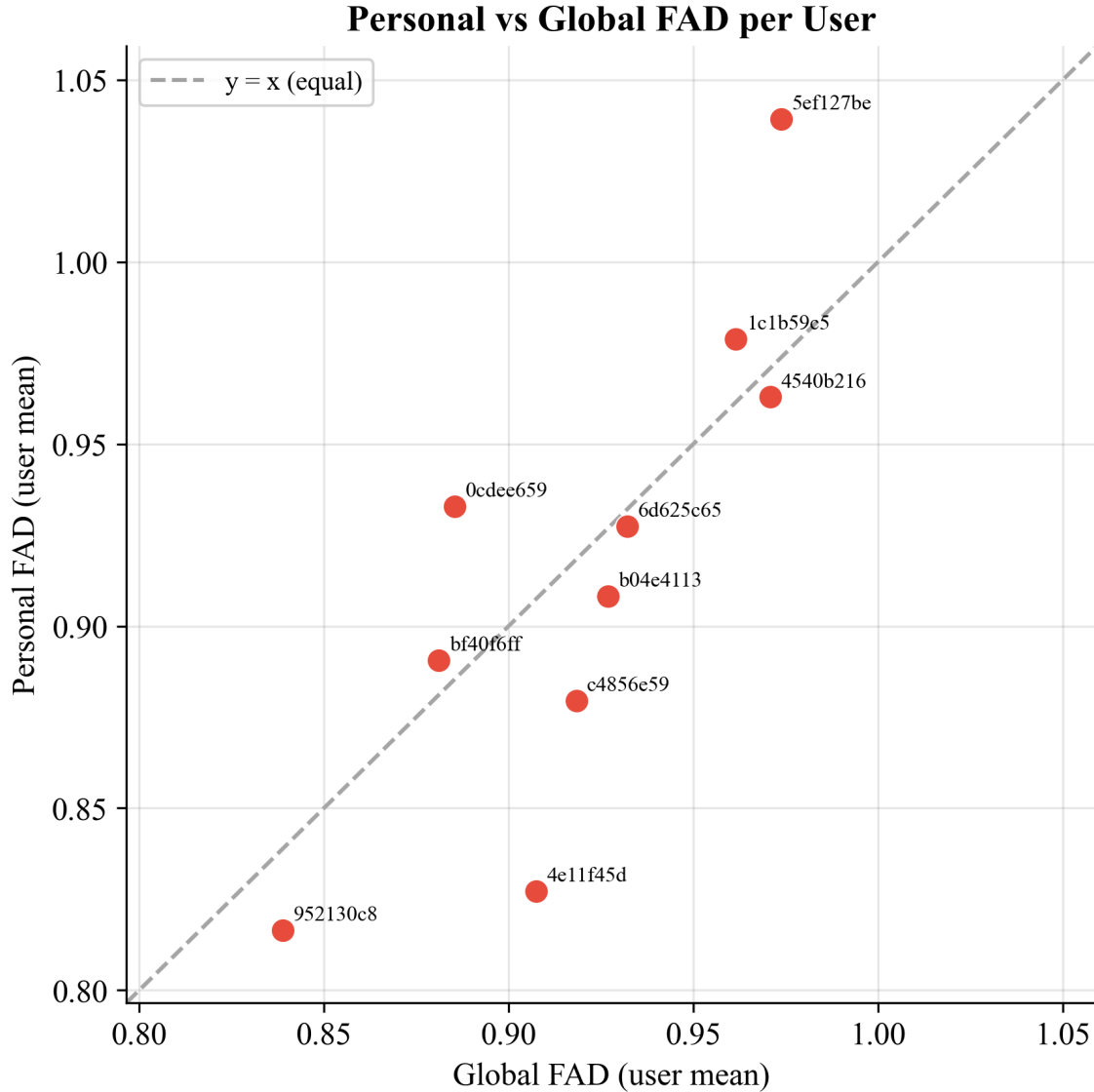


**Figure 3.** Mean CLAP text-to-audio score across the three conditions.

#### 4.1.2 Per-user Analysis

Figure 4 presents a scatter plot that compares per-user mean FAD scores for personal versus global augmentation methods. Each point represents one user's mean FAD across all vague prompts. Points below the diagonal indicate users whose personal augmentation resulted in lower FAD than global. The majority of points cluster near the diagonal. This is consistent with Figure 2, where no significant overall difference was observed between personal and global augmentation. However, individual variation is visible and some

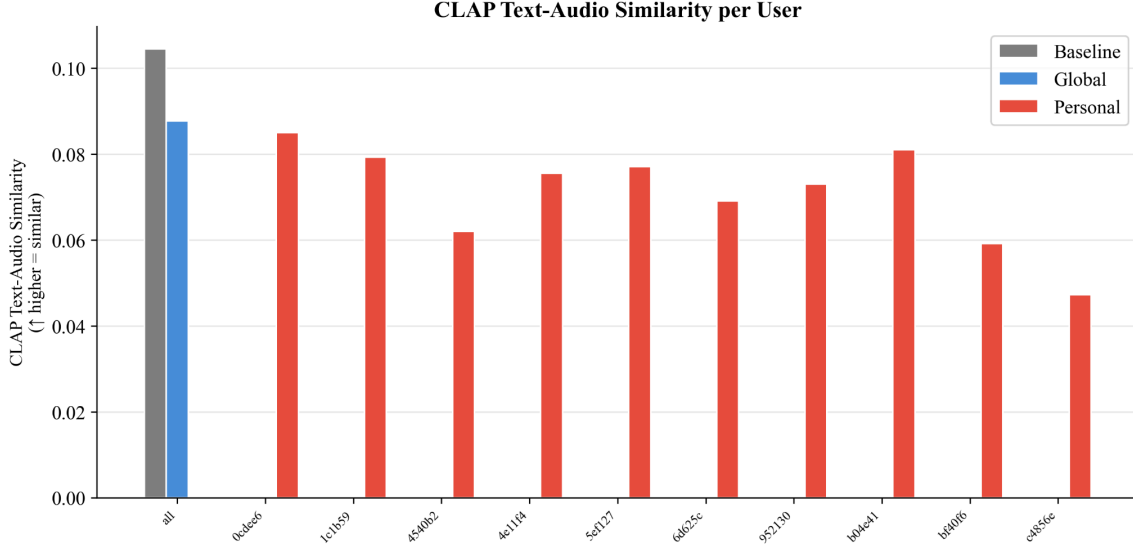
users benefit substantially from personal augmentation while others do not. This suggests that the effectiveness of personalization may depend on user-level factors such as the diversity of their listening history or the specificity of their tag co-occurrence patterns.



**Figure 4.** Scatter plot of per-user mean FAD for personal versus global augmentation.

Figure 5 presents CLAP text-audio similarity scores per user. Since the baseline and global conditions use the same prompt for all users, they produce a single score each (0.105 and 0.088 respectively). Personal augmentation scores vary across users, ranging

from 0.047 to 0.085. All personal scores fall below the baseline, consistent with the overall trend observed in Figure 3.

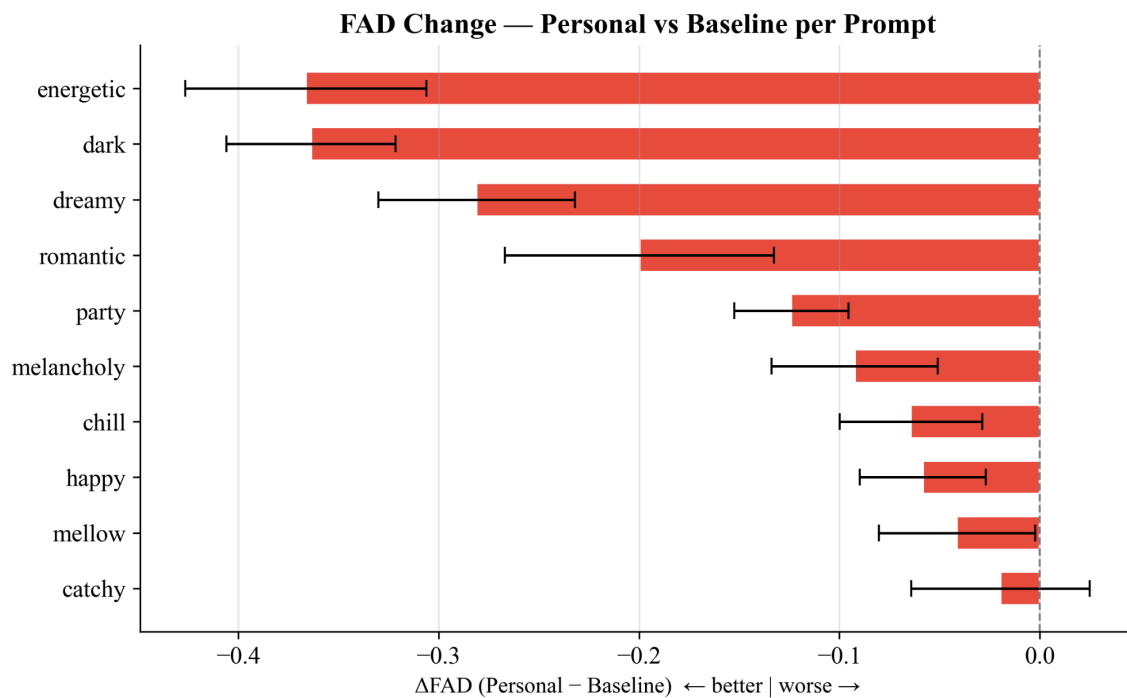


**Figure 5.** *CLAP text-audio similarity scores per user.*

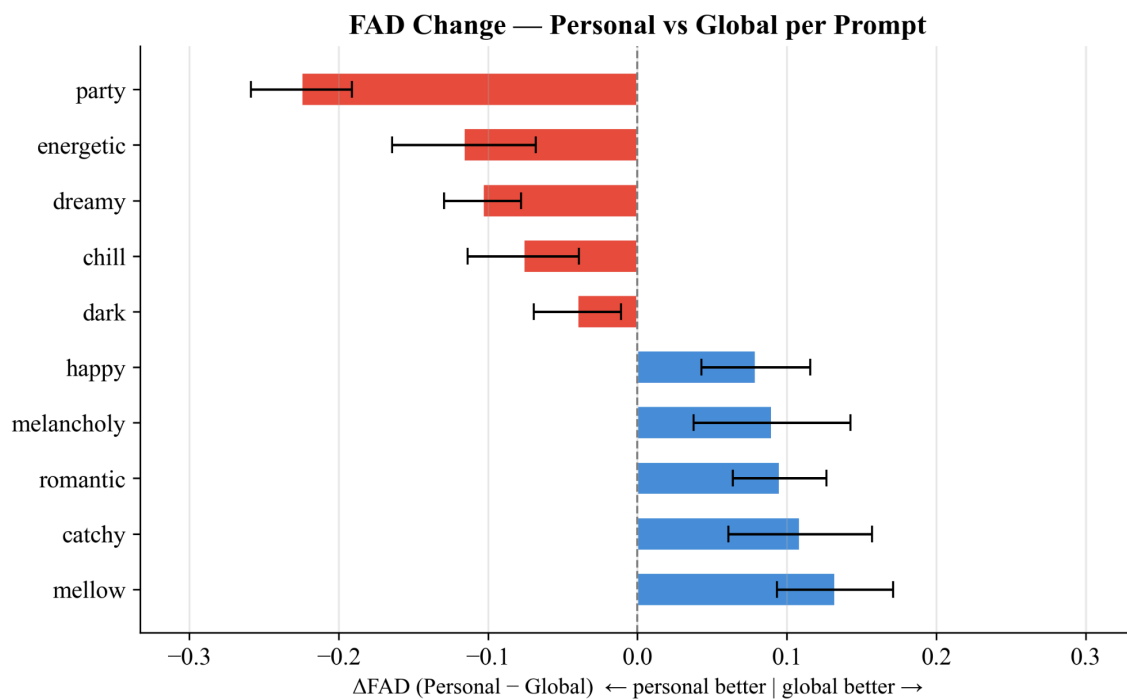
#### 4.1.3 Per-Prompt Analysis

Figure 6 presents the FAD change from the unaugmented baseline for personal augmentation per prompt. Negative values indicate a lower FAD, indicating greater distributional similarity to the user’s reference audio. The biggest differences were observed for "energetic" ( $\Delta=-0.37$ ), "dark" ( $\Delta=-0.36$ ), and "dreamy" ( $\Delta=-0.28$ ). The smallest differences were for "mellow" ( $\Delta=+0.04$ ) and "chill" ( $\Delta=+0.06$ ).

Figure 7 presents the FAD difference between personal and global augmentation per prompt. Red bars extending left indicate prompts where personal augmentation produced lower FAD than global, while blue bars extending right indicate prompts where global produced lower FAD than personal. The largest personal advantage is observed for “party” ( $\Delta=-0.22$ ), while the largest global advantage is for “mellow” ( $\Delta=+0.13$ ).

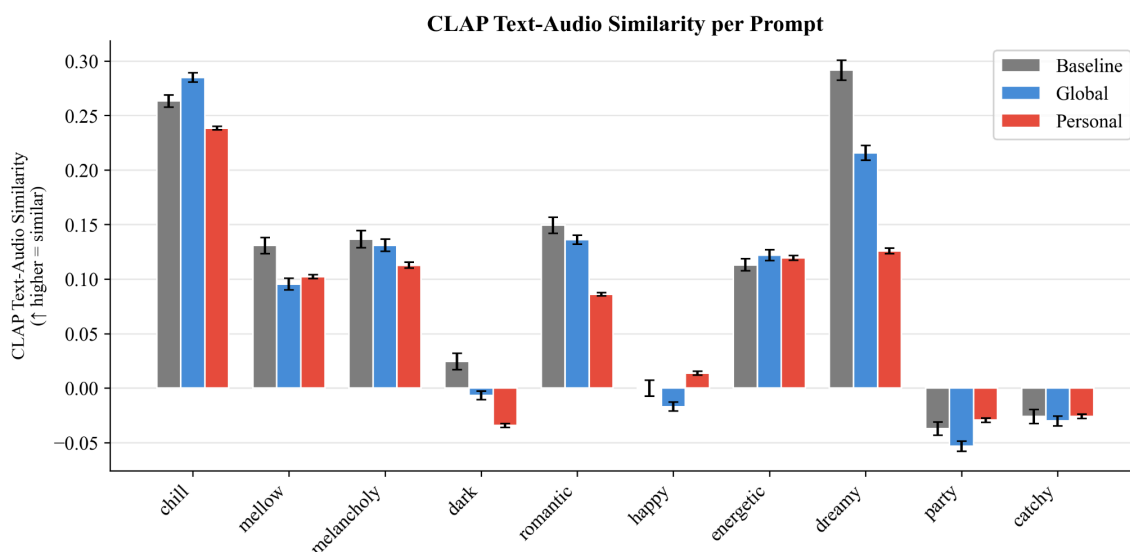


**Figure 6.** *FAD change from the unaugmented baseline for personal augmentation per prompt.*



**Figure 7.** *FAD difference between personal and global augmentation per-prompt.*

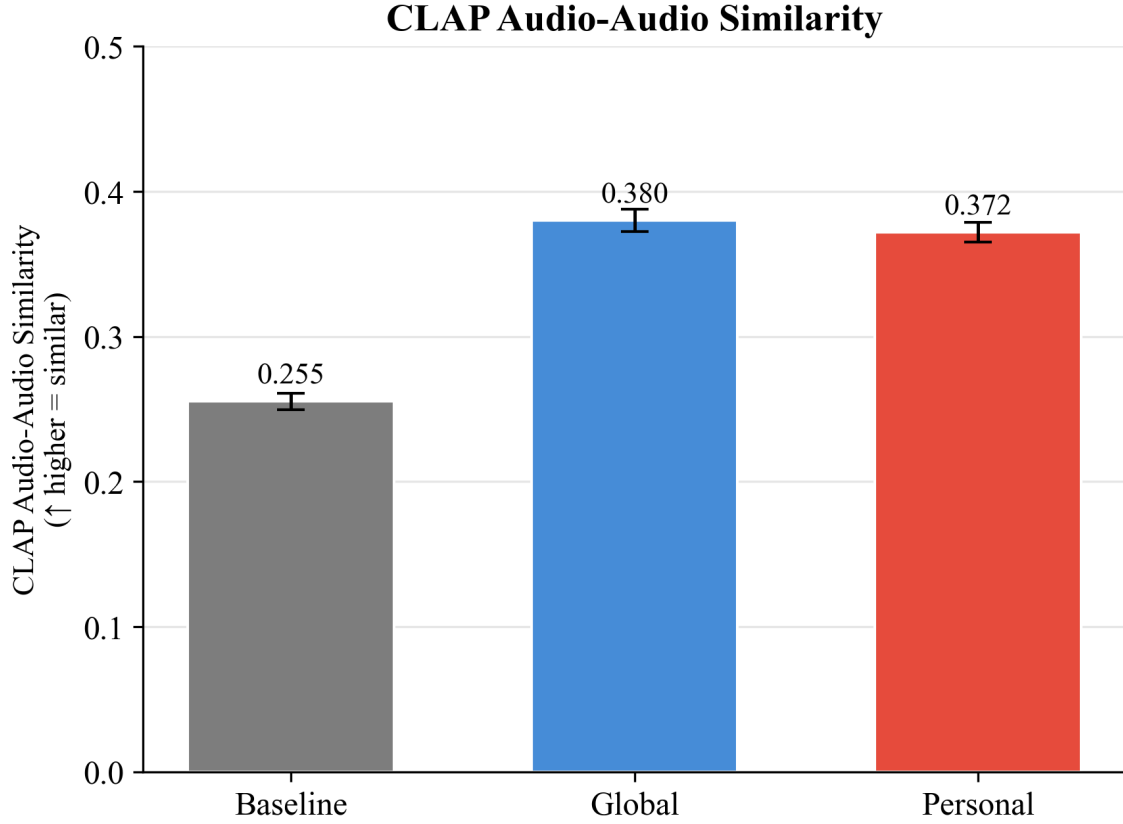
Figure 8 shows CLAP text-audio similarity scores per prompt across the three conditions. Vague prompts like “chill” and “dreamy” show the highest similarity. Prompts like “dark”, “happy”, “party”, and “catchy” show near-zero or negative scores. For most prompts, the three conditions produce similar CLAP scores. No single condition consistently scores higher or lower. This supports the finding from figure 3 that augmentation does not significantly affect proxy concept retention at the per-prompt level.



**Figure 8.** CLAP text-audio similarity scores per prompt across the three conditions.

#### 4.1.4 Complementary Results

Figure 9 presents the CLAP audio-audio similarity scores across the three conditions. Higher values indicate greater similarity between generated audio and the user’s held-out reference audio in the CLAP embedding space. Both personal (0.372) and global (0.380) significantly outperformed the baseline (0.255,  $p < 0.0001$ ), with no significant difference between personal and global ( $p = 0.34$ ). These results corroborate the FAD findings, confirming that both augmentation strategies produce audio closer to user preferences than the baseline.



**Figure 9.** *CLAP audio-audio similarity scores across the three conditions.*

Table 2 presents user ranking scores across the three conditions. Hit Rate@1 was identical across all conditions (0.120). This shows that no condition consistently ranked the correct user first. However, at Hit Rate@3, personal augmentation (0.434) shows improvement upon both baseline and global (0.361). Personal augmentation also achieved a higher MRR (0.347 vs 0.328) and a lower mean rank (4.651 vs 4.880). These results suggest that personal augmentation produces audio that is more identifiable with the correct user’s reference set library compared to the other conditions.

Full statistical test results can be found in the Appendix A.

| Metric                                    | Unaugmented<br>(Baseline) | Global<br>Augmentation | Personal<br>Augmentation |
|-------------------------------------------|---------------------------|------------------------|--------------------------|
| Hit Rate @ 1                              | 0.120                     | 0.120                  | 0.120                    |
| Hit Rate @ 3                              | 0.361                     | 0.361                  | <b>0.434</b>             |
| Mean Reciprocal Rank (MRR)                | 0.328                     | 0.328                  | <b>0.347</b>             |
| Mean Rank ( $\downarrow$ lower is better) | 4.880                     | 4.880                  | <b>4.651</b>             |

**Table 2.** *User ranking metrics across the three conditions.*

## 4.2 Discussion

The main research question of this study is whether augmenting vague text prompts with listening-history-derived metadata improves personalization in text-to-music generation. The experimental results show that both augmentation methods helped MusicGen generate music that is distributionally closer to each user's held-out reference set compared to the unaugmented baseline. However, this alone does not prove personalization. It only shows that adding more descriptive tags to a vague prompt helps MusicGen produce audio closer to the reference set.

Sub-RQ2 directly tests personalization by comparing personal augmentation against global augmentation. Neither the mean FAD comparison nor the per-user analysis showed a significant difference between the two conditions. The per-user scatter plot in Section 4.1.2 confirms this, with most users clustering near the diagonal. Complementary CLAP audio-to-audio and user ranking results also support this finding. It is worth noting that the prompt length confound does not affect this comparison, as both augmented conditions use prompts of similar length. The minimal difference between conditions suggests that at the aggregate level, the specific content of the co-occurring tags matters less than the act of augmentation itself.

Qualitative inspection of the augmented prompts reveals that personalized prompts do contain varied and user-specific tags, suggesting that the co-occurrence method is capable

of capturing meaningful individual differences. The gap between this qualitative observation and the quantitative results may indicate that the per-user evaluation data was not sufficient to detect a significant effect, or that user-specific tags do not consistently translate into measurably different audio outputs through MusicGen.

Sub-RQ3 looked at whether augmented prompts preserve alignment with the original vague concept. Figures 3, 5, and 8 show a downward trend in CLAP text-audio similarity for augmented conditions, but statistical analysis did not show a significant change. This trend can be attributed to augmented prompts distributing MusicGen's attention across multiple tags rather than focusing solely on the vague term. The non-significant differences suggest that proxy concept alignment is largely preserved despite augmentation.

Sub-RQ4 asked whether specific prompts perform better under specific conditions, and this is where the most important finding of this thesis emerges. Personalization was significantly more effective for "party" ( $p < 0.001$ ,  $d = 2.21$ ) and "dreamy" ( $p = 0.007$ ,  $d = 1.64$ ), with near-significant trends for "energetic" ( $p = 0.061$ ) and "chill" ( $p = 0.070$ ). These are experiential and activity-oriented terms whose musical interpretation can vary substantially across listeners. One user's "party" music might be hip-hop while another's might be electronic dance music. For these terms, the user's own listening history provides more relevant co-occurrence information than the collective average. Conversely, global augmentation significantly outperformed personal for "mellow" ( $p = 0.008$ ) and "romantic" ( $p = 0.038$ ), suggesting that these prompts have more consistent musical associations across listeners where the larger global data pool produces more stable tag associations. This pattern was also supported by the supplementary user ranking analysis, which showed higher hit rates for personal augmentation on the same prompts where FAD favored personal.

## 5. CONCLUSIONS

This thesis proposed a prompt augmentation method for personalizing vague text-to-music generation prompts using PMI-weighted tag co-occurrence patterns derived from listening history metadata. The results show that both global and personal augmentation improve alignment with user listening preferences without sacrificing concept retention. However, personal augmentation did not significantly outperform global augmentation overall, suggesting that the added value of user-specific personalization depends on the prompt term. Personalization shows promise for specific prompt types where individual taste varies most.

### 5.1 Limitations

Several limitations should be acknowledged. Listening histories represent implicit feedback and do not reflect explicit preference (McFee et al., 2012). A user having listened to a song does not guarantee preference for it. This means that all evaluation metrics in this study serve as proxies for personalization rather than direct measures of user satisfaction. Since the users in the Taste Profile dataset are completely anonymized, a user study with the original users cannot be conducted. Further complementary user studies are required to validate these proxy results as evidence of true personalization.

The global co-occurrence matrix is computed from all 50 users' data, providing a richer statistical basis than any single user's 400-song history. Some user-prompt pairs had sparse tag coverage, producing less informative personal augmentations. These sources of noise affect personal augmentation more than global, since the global condition benefits from averaging across many users.

The improvement of augmented conditions over the baseline may be partially attributable to increased prompt length rather than co-occurrence content. The comparison between personal and global augmentation mitigates this confound, as both conditions use prompts of similar length. Seventeen out of 100 user-prompt pairs were excluded due to insufficient reference data, primarily affecting "romantic", "energetic", and "dark".

The reference audio was sourced from YouTube and may not perfectly match the original recordings. The co-occurrence patterns reflect biases in the Last.fm user community and may not generalize broadly. MusicGen generates 10-second clips, which may not capture the full musical character implied by a vague term. While FAD correlates with human quality judgments (Kilgour et al., 2019), the perceptual significance of the observed differences remains unclear without human evaluation. Some per-user per-prompt reference sets remain small after chunking, which may affect FAD stability.

## ***5.2 Future Work***

This work can be extended in several directions. A larger and more carefully controlled dataset with a cleaner development and reference split would strengthen the evaluation. Design decisions such as play count weighting, prompt length, and converting comma-separated tags to natural language prompts could be explored. Other music generation models such as Stable Audio or MusicLDM could be tested, and a benchmark for personalized music generation could be developed. Most importantly, a human evaluation framework through listening tests would be crucial to validate whether the objective improvements observed in this study correspond to meaningful perceptual differences for listeners. Designing such a study was out of the scope of this thesis but remains an essential next step.

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## Literature Review

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## Method

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## APPENDIX A: Statistical Test Results

This appendix reports the statistical test results for the main condition-level comparisons and the per-prompt FAD comparisons. For FAD, negative  $\Delta$  values indicate that the first condition achieved lower FAD, which means better distributional similarity to the reference set. Significance is marked as  $p < .05$  (\*),  $p < .01$  (\*\*),  $p < .001$  (\*\*\*), and ns for non-significant results.

| Comparison                | $\Delta$ | t      | p        | Cohen's d | n  | sig |
|---------------------------|----------|--------|----------|-----------|----|-----|
| Personal vs Baseline FAD  | -0.136   | -7.585 | 0.000000 | -0.838    | 83 | *** |
| Personal vs Global FAD    | -0.002   | -0.134 | 0.893419 | -0.015    | 83 | ns  |
| Global vs Baseline FAD    | -0.134   | -8.630 | 0.000000 | -0.953    | 83 | *** |
| Personal vs Baseline CLAP | -0.034   | -1.981 | 0.078927 | -0.660    | 10 | ns  |
| Personal vs Global CLAP   | -0.017   | -1.431 | 0.186335 | -0.477    | 10 | ns  |
| Global vs Baseline CLAP   | -0.017   | -1.975 | 0.079734 | -0.658    | 10 | ns  |

**Table A1.** Overall statistical test results for FAD and CLAP text-audio similarity.

Table A2 reports the per-prompt FAD comparisons between personal and global augmentation. Negative  $\Delta$  values indicate prompts where personal augmentation achieved lower FAD than global augmentation, while positive  $\Delta$  values indicate prompts where global augmentation achieved lower FAD than personal augmentation.

| Prompt | Personal | Global | $\Delta$ | p      | sig |
|--------|----------|--------|----------|--------|-----|
| chill  | 1.0      | 1.0    | -0.1     | 0.0701 | ns  |
| mellow | 0.9      | 0.8    | +0.1     | 0.0078 | **  |

|                   |     |     |      |        |     |
|-------------------|-----|-----|------|--------|-----|
| <b>melancholy</b> | 0.9 | 0.8 | +0.1 | 0.1197 | ns  |
| <b>dark</b>       | 0.9 | 0.9 | 0.0  | 0.2263 | ns  |
| <b>romantic</b>   | 1.1 | 1.0 | +0.1 | 0.0384 | *   |
| <b>happy</b>      | 0.9 | 0.8 | -0.1 | 0.0573 | ns  |
| <b>energetic</b>  | 0.9 | 1.0 | -0.1 | 0.0607 | ns  |
| <b>dreamy</b>     | 0.9 | 1.0 | -0.1 | 0.0070 | **  |
| <b>party</b>      | 0.9 | 1.1 | -0.2 | 0.0001 | *** |
| <b>catchy</b>     | 0.9 | 0.8 | +0.1 | 0.0528 | ns  |

**Table A2.** *Per-prompt FAD comparison between personal and global augmentation.*